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Research Insights

Promotion of chain growth and suppression of chain transfer by 2-methoxy group in diarylmethyl moiety in pyridine-imine nickel- and palladium-catalyzed ethylene (co)polymerization

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ABSTRACT

Diarylmethyl substituents are commonly used to prevent chain transfer in late-transition metal-catalyzed ethylene polymerization. However, this substituent is not effective in inhibiting chain transfer in the pyridine-imine system. In this study, we introduced substituents at the 2-position of the diarylmethyl moiety to synthesize high molecular weight polyethylene in the pyridine-imine system. Using this strategy, we synthesized two pyridine-imine ligands with either ethyl or methoxy at the 2-position diarylmethyl moiety, along with their corresponding nickel and palladium complexes. These complexes were then utilized for ethylene (co)polymerization. The catalysts with OMe exhibited significantly higher polymerization activity (an order of magnitude, 10^5 vs. 10^4 g/(mol Ni·h)) compared to the corresponding Et catalysts. In addition, it was found that polyethylene produced from nickel and palladium catalysts with OMe has higher molecular weights than that produced from ethyl nickel and palladium catalysts. The nickel system produces polyethylene with a molecular weight up to an order of magnitude higher (an order of magnitude, 13.0–13.6 vs. 3.1–4.0 kg/mol). Similar results were also observed in the palladium-catalyzed copolymerization of ethylene with MA. Notably, the palladium catalyst with OMe has a significantly higher capacity for MA insertion (three times, 1.68–4.74 vs. 5.00–11.73 mol%) than the ethyl palladium catalyst, despite their similar spatial steric structures. The observed phenomenon may be attributed to the weak interaction between OMe and the metal center. In addition, ^{13}C NMR analysis showed that the palladium-catalyzed polyethylene and E-MA copolymers are hyperbranched polymers, with the polar groups located at the ends of the branches.

1. Introduction

The nickel and palladium α-diimine catalysts, specifically the pyridine-imine variants, are capable of catalyzing the polymerization of ethylene through the chain-walking polymerization mechanism, resulting in the production of branched polyethylene. Additionally, the palladium catalysts can catalyze the copolymerization of ethylene with acrylate monomers, producing polar functionalized polyolefins [1–19]. In addition to the aforementioned similarities, low molecular weight polyethylene and polar copolymers are frequently obtained as catalysts with one-sided steric hindrance [7–17]. This feature gives the impression that the pyridine-imine nickel and palladium system is only capable of catalyzing ethylene (co)oligomerization. Some strategies involving bulky *o*-aryl substituents can significantly suppress chain transfer in α-diimine systems. However, these same strategies may not have the same effect in pyridine-imine systems and may even promote chain transfer instead. For instance, the o-aryl diarylmethyl and aryl substituents effectively suppress chain transfer in α-diimine systems, resulting in the synthesis of higher molecular weight polyethylene [20–28]. Conversely, in the corresponding pyridine-imine systems, they promote chain transfer, leading to the synthesis of lower molecular weight ethylene oligomers (Chart 1**a-b**) [29–33]. Our additional research has shown that substituting benzothiophene in the diarylmethyl α-diimine system can effectively reduce chain transfer and chain walking, resulting in the production of low-branched ultrahigh molecular weight polyethylene (UHMWPE) [34]. Conversely, in the corresponding pyridine-imine system, this substitution has the opposite effect and further inhibits chain growth, leading to very low activity (Chart 1**c**)

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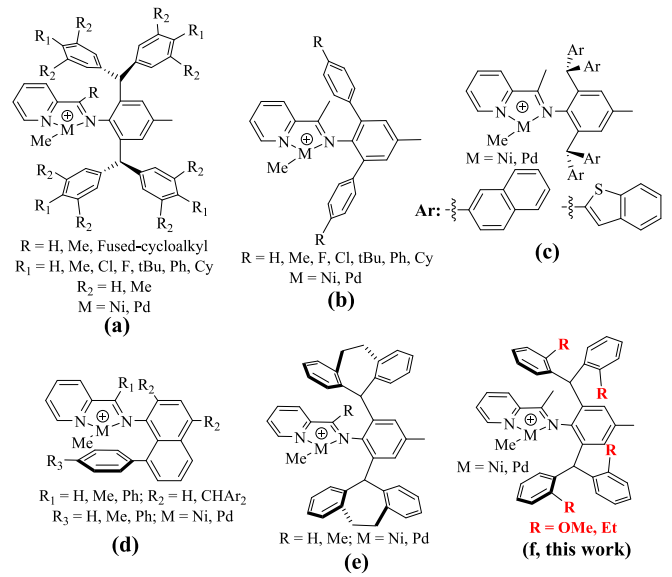


Chart 1. Recently published pyridine-imine Ni(II) and Pd(II) catalysts (**a**-**e**), and our current work (**f**).

[35]. Our group has recently made progress in synthesizing high molecular weight polyethylene and copolymers by suppressing chain transfer in pyridine-imine systems using some interesting strategies. For example, the 8-aryl-naphthyl substituent (Chart 1**d**) and the rotation-restricted dibenzosuberyl group (Chart 1**e**) have been shown to effectively suppress chain transfer, resulting in high molecular weight polyethylene and copolymers during pyridine-imine nickel and palladium catalyzed ethylene polymerization [18,19,36–38]. In this study, two types of substituents are used to restrict the rotation of the aryl group in the diarylmethyl group of the pyridine-imine system to regulate the olefin polymerization process (Chart 1**f**). Methoxy is found to suppress chain transfer and promote chain growth more effectively than the corresponding ethyl group in this system, resulting in significantly higher molecular weight polymers with much greater activity in ethylene polymerization.

Table 1
Ethylene Polymerization with Nickel Complexes.[a]

Entry	Complex	T (°C)	Yield (mg)	Act.[b]	M_n[c]	M_w/M_n[c]	*brs*[d]
1	**Ni1**	30	22	0.22	3.1	1.64	49
2	**Ni1**	50	14	0.14	3.3	1.55	43
3	**Ni1**	70	12	0.12	4.0	1.64	60
4	**Ni2**	30	340	3.40	13.0	1.75	41
5	**Ni2**	50	246	2.46	13.2	2.95	45
6	**Ni2**	70	242	2.42	13.6	2.15	60

[a] Conditions: Ni(II) complexes (2 μmol), 200 eq. Et_2AlCl, 1 mL of CH_2Cl_2, 20 mL toluene, polymerization time (30 min), 6 atm.
[b] Activity (Act.) = 10^5 g/(mol Ni·h).
[c] Molecular weight (kg·mol^{-1}) was determined by GPC in THF at 40 °C using polystyrene standards.
[d] brs = branches per 1000 carbons, determined by ^{1}H NMR spectroscopy, brs = 1000 × 2(I_{CH3})/3(I_{CH2+CH} + I_{CH3}).

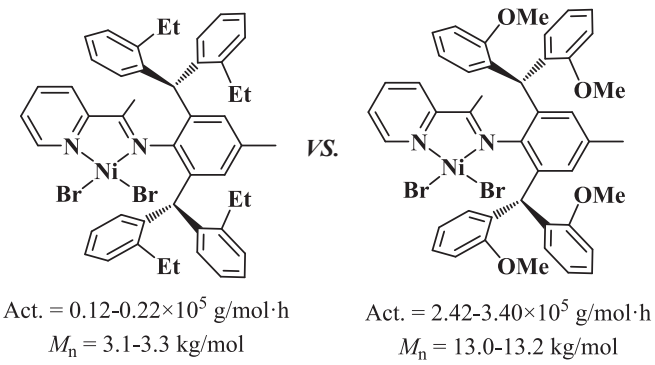
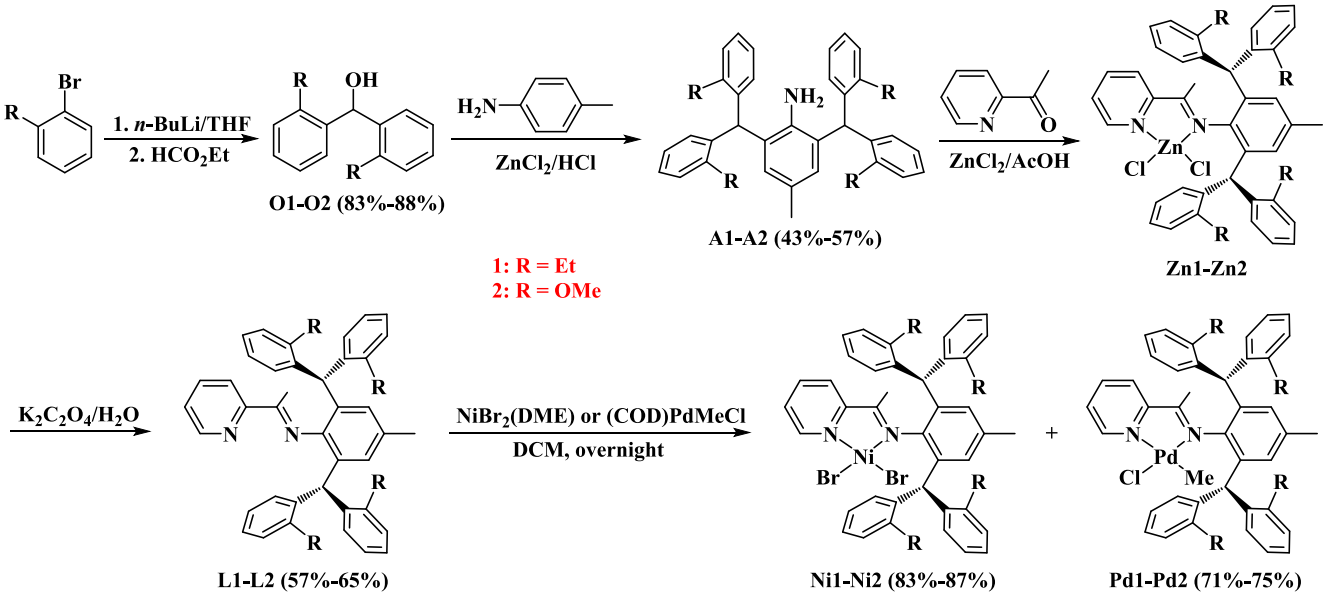


Fig. 1. Comparisons on activity and molecular weight of the polyethylenes produced with **Ni1** and **Ni2** at 6 atm.



Scheme 1. Synthesis of pyridine-imine ligands and the corresponding nickel and palladium complexes.

Table 2
Ethylene polymerization with palladium complexes.[a]

Entry	Complex	T (°C)	Yield (g)	Act.[b]	M_n[c]	M_w/M_n[c]	*brs*[d]
1	**Pd1**	30	0.16	0.53	1.6	1.17	45
2	**Pd1**	50	0.11	0.37	1.4	1.09	44
3	**Pd1**	70	0.03	0.10	1.1	1.10	60
4	**Pd2**	30	1.31	4.37	2.8	1.35	54
5	**Pd2**	50	2.44	8.13	2.2	1.15	53
6	**Pd2**	70	1.47	4.90	1.5	1.21	65

[a] Conditions: Pd(II) complexes (10 μmol), NaBArF (2.0 equiv.), ethylene (4 atm), DCM (40 mL), polymerization time (3 h).
[b] Activity is in unit of 10^4 g·mol^{-1}·h^{-1}.
[c] Molecular weights (kg·mol^{-1}) was determined by GPC in THF at 40 °C using polystyrene standards.
[d] *brs* = branches per 1000C carbons, determined by ^{1}H NMR spectroscopy, *brs* = 1000 × 2(I_{CH3})/3(I_{CH2+CH} + I_{CH3}).

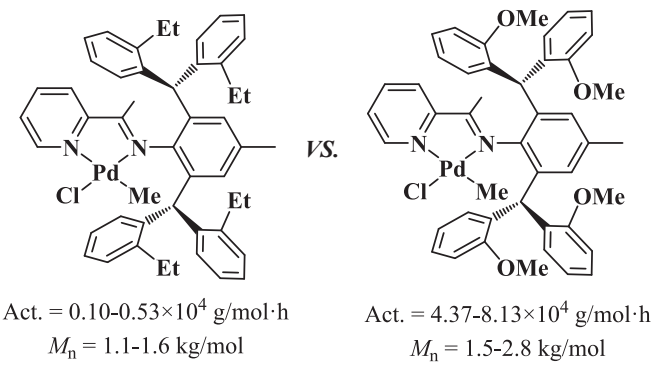


Fig. 2. Comparisons on activity and molecular weight of the polyethylenes produced with **Pd1** and **Pd2** at 4 atm.

2. Results and discussion

2.1. Pyridine-imine nickel and palladium complexes synthesis

Diarylmethylanilines **A1-A2**, with ethyl or methoxy at the 2-position aryl moiety, were synthesized by alkylating *p*-toluidine with the corresponding diarylmethanols using Lucas' reagent, following our previously reported method [22,23]. The diarylmethanol feedstocks were prepared through the nucleophilic addition of aryl lithium, which was generated in situ, with ethyl formate. The pyridine-imine ligands **L1-2** can be prepared through the condensation of 2-acetylpyridine with **A1-2** using the zinc chloride template method (Scheme 1). The diarylmethylanilines **A1-2** and ligands **L1-2** were obtained in moderate yields (43–65 %), which were lower than the yields reported for the corresponding unsubstituted analogs [29,31]. This lower yield may be attributed to the presence of the 2-substituent group. The ^{1}H and ^{13}C NMR spectra and high resolution mass spectrometry (HRMS) were used to confirm the diarylmethylanilines and ligands. The pyridine-imine ligands were reacted with catalyst precursors ($NiBr_2$(DME) and (COD) PdMeCl) in DCM solvent to synthesize the nickel and palladium complexes. After an overnight reaction, most of the DCM was removed and diethyl ether was added to precipitate the nickel and palladium complexes. **Ni1-2** and **Pd1-2** complexes were analyzed for composition and molecular mass using elemental analysis (C, H, N mode) and MALDI-TOF mass spectrometry. In addition, **Pd1-2** were identified using ^{1}H and ^{13}C NMR spectra.

2.2. Ethylene polymerization catalyzed by nickel complexes.

At 200 equivalents of diethylaluminium chloride activation, **Ni2** with OMe demonstrates polymerization activity that is one order of magnitude higher than **Ni1** with Et (Table 1, Fig. 1). The activity of both catalysts decreased with an increase in temperature. Notably, the molecular weight of the polyethylene produced by **Ni2** is also an order of magnitude higher than that of **Ni1** (Fig. 1). The rise in temperature resulted in an increase in the molecular weight of both catalysts. The reason for this phenomenon has not been definitively explained, in general, the molecular weight of polyethylene produced is predominantly influenced by the ratio of chain growth in comparison to chain transfer. Typically, an elevation in temperature expedites the chain transfer reaction [21–23,31–33]. Paradoxically, despite the nickel catalyst's diminishing activity with rising temperature in our study, the molecular weight of the polyethylene generated actually increases. We postulate that while temperature briefly boosts chain growth, the catalyst's instability at higher temperatures leads to a decline in its effective activity. Consequently, the molecular weight enhancement could be

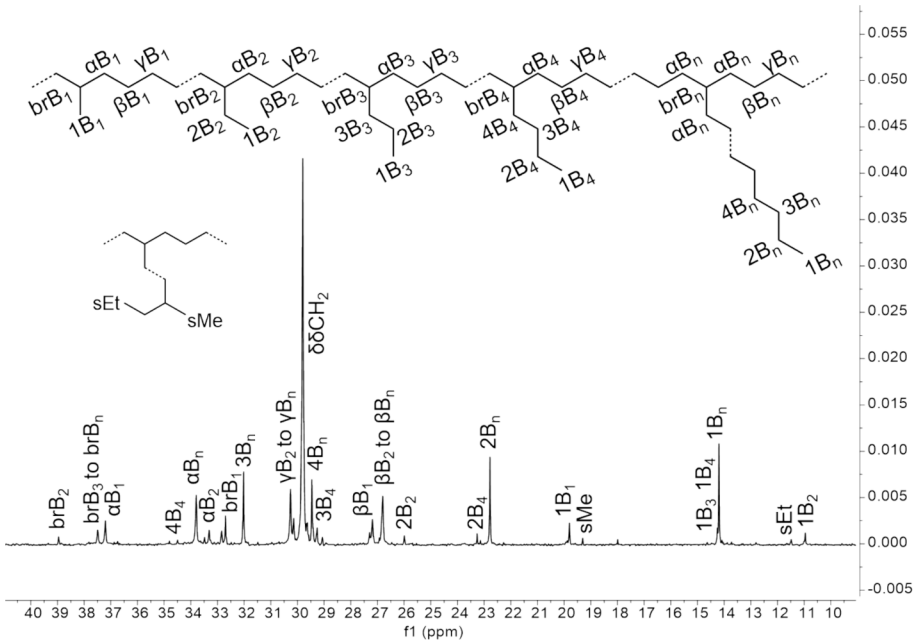


Fig. 3. Branching type assignments of the polyethylene from entry 3, Table 2 using ^{13}C NMR spectrum analysis based on ref. 39–41.

Table 3
Copolymerization of Ethylene and MA with Palladium Complexes.[a]

Entry	Complex	[MA]	Yield (g)	TOF[b] E/MA	X_{MA} (mol %)[c]	M_n [c]	M_w/M_n[c]	N[d]	brs[e]
1	**Pd1**	1	0.13	18.38/0.31	1.68	1.6	1.08	0.93	85
2	**Pd1**	2	0.11	16.34/0.71	4.74	1.5	1.05	2.31	99
3	**Pd2**	1	3.50	448.4/23.60	5.00	2.2	1.18	2.87	79
4	**Pd2**	2	1.18	124.8/16.57	11.73	2.2	1.18	7.41	89

[a] Conditions: Pd(II) complexes (20 μmol), NaBArF (2.0 equiv.), total volume of CH_2Cl_2 and MA (20 mL), polymerization time (12 h), ethylene (4 atm), polymerization temperature (40 °C).
[b] TOF is in unit of h^{-1}.
[c] X_{MA} = Incorporation of MA.
[d] N = Average molar number of functional groups per molar copolymer chain, calculated from ^{1}H NMR.
[e] brs = Number of branches per 1000C, as determined by ^{1}H NMR spectroscopy. The functional branches were also added to the total branches.

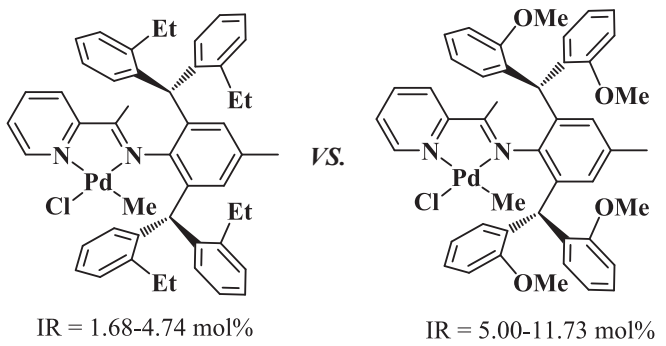


Fig. 4. Comparisons on the MA insertion ratio of E-MA copolymers produced with **Pd1** and **Pd2** at 4 atm.

attributed to an augmented ratio of fleeting activity to chain transfer. Additionally, the polyethylene produced by this system has a moderate degree of branching.

2.3. Ethylene (co)polymerization catalyzed by palladium complexes

Under the activation of two equivalents of tetrakis(3,5-bis (trifluoromethyl)phenyl)borate (NaBArF), complex **Pd2** with OMe exhibits ethylene polymerization activity that is one order of magnitude higher than **Pd1** with Et (Table 2, Fig. 2). The activity of **Pd1** decreased with an increase in temperature, while **Pd2** showed a significant improvement from 30 to 50 °C and then decreased at 70 °C. Similar to the nickel system, polyethylene catalyzed by **Pd2** has a higher molecular weight than that of **Pd1**, but the difference is relatively small (Fig. 2). The increase in activity from 30 to 50 °C can be attributed to the weak interaction of OMe with the metal center and its ability to prevent the palladium catalyst from undergoing the inactivation reaction through C–H activation. In addition, the polyethylene produced by this system is moderately branched, which is significantly less than other pyridine-imine palladium systems [18,19,36–38]. This may be due to its rotationally restricted steric structure. Further analysis using ^{13}C NMR revealed that the polyethylenes contain various types of branching, including methyl, ethyl, propyl, C_{3+} branching, and branched-on-branched microstructures [39–41]. Long-chain (C_{3+}) branching accounts for the majority of all branching (Fig. 3).

Recent studies have shown that pyridine-imine palladium catalysts efficiently catalyze the copolymerization of ethylene with methyl acrylate (MA) [18,19,31–33]. **Pd2** demonstrates significantly higher copolymerization activity (11–27 times) than **Pd1** in the copolymerization of ethylene with MA, which is consistent with its performance in ethylene homopolymerizations, where the difference is even more pronounced (Table 3). Additionally, **Pd2** produces higher molecular weights than **Pd1**. Interestingly, **Pd2** has a significantly higher capacity for MA insertion than **Pd1** (Fig. 4), despite their similar spatial steric structures.

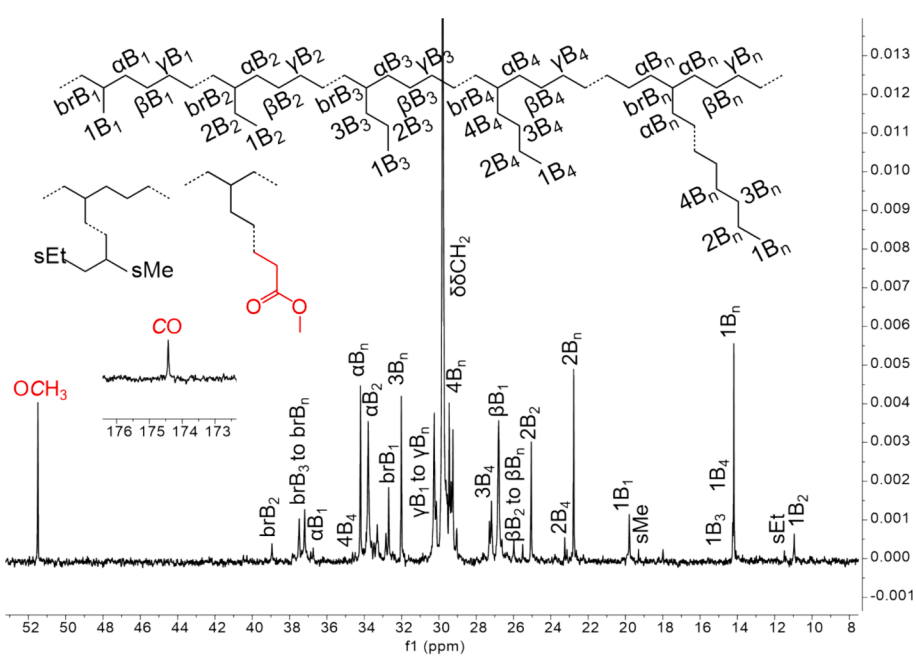


Fig. 5. Branching type assignments of the E-MA copolymer from entry 4, Table 3 using ^{13}C NMR spectrum analysis based on ref. 18–19 and 39–41.

This could be due to the weak interaction between methoxy and the metal center, which promotes the opening of the ester chelate ring after the insertion of MA. As anticipated, higher concentrations of MA resulted in decreased copolymerization activity and increased MA insertion ratio for both catalysts (Table 3). The copolymers exhibit a high degree of branching and, on average, several MA insertions per copolymer chain (Table 3). The analysis of the ^{13}C NMR spectrum of the copolymer from Table 3, entry 4 indicates that the copolymer is rich in branched types, similar to the polyethylenes obtained by the palladium complexes (Fig. 5) [18,19,39–41]. Additionally, the ester-functionalized long-chain branching was also observed.

3. Conclusions

In summary, two diarylmethyl pyridine-imine ligands with ethyl and methoxy at the 2-position diarylmethyl moiety and the corresponding nickel and palladium complexes were synthesized and used for ethylene (co)polymerization. **Ni2** with OMe exhibited an order of magnitude higher polymerization activity and produced polyethylene with an order of magnitude higher molecular weight than **Ni1** with Et. Similar results were also observed in the palladium-catalyzed polymerization of ethylene and copolymerization of ethylene with MA. Notably, **Pd2** has a significantly higher capacity (3 times) for MA insertion than **Pd1**, despite their similar spatial steric structures. ^{13}C NMR analysis revealed that the palladium-catalyzed polyethylene and E-MA copolymers are hyperbranched polymers, with the polar groups located at the ends of the branches. The observed phenomenon may be attributed to the weak interaction between OMe and the metal center.

CRediT authorship contribution statement

Qi Cai: Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation. **Shengyu Dai:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Full experimental details for the synthetic procedures, materials, analytical methods, NMR and mass spectra of synthetic compounds, NMR and GPC curves of polymers and copolymers samples. Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.113132.

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